TECHNICAL DILIGENCE REVIEW - RECORD 005

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-005

1. Scope, targets and the basis of the capacity assessment

2025-10-24 is the evidence cut-off for report TDR-2025-17, whose subject is the capacity position, load-test evidence and latency behaviour of the platform at fleet scale. This report is one of a series of technical reports; this one takes the capacity position, load-test evidence and latency behaviour of the platform at fleet scale and reports against the evidence available by 2025-10-24. Work on report TDR-2025-17 ran over 7 weeks and included 17 interviews and read-only access to the engineering repositories. J. Ferreiro led the work behind this report, combining document review, 17 interviews and hands-on inspection of running systems. The capacity position, load-test evidence and latency behaviour of the platform at fleet scale is a wide field, and the present report is explicit about which parts of it were reached.

21 services were examined in detail for the present report, chosen for their position in the dependency map rather than at random. TDR-2025-17 covers 21 of 34 services; the untested remainder is listed so a reader can see what was not reached. Open items in the report at TDR-2025-17 are stated with what would close each one, so a post-close plan can be costed against them. The report at TDR-2025-17 closes with 5 open items, listed with owner, evidence and the reviewer's view of whether the committed date is realistic. Report TDR-2025-17 contains 5 open items, and a reader should assume that number will move rather than settle.

The working method behind this report was to start from the incident record and work back to the design, rather than the other way round. Access for report TDR-2025-17 was sufficient for the conclusions drawn; where it was not, the limitation is stated at the point it bites. This report suggests one governance change rather than a price adjustment: a quarterly technical review with the lead investor's observer present. 6 items in TDR-2025-17 are post-close engineering work rather than transaction issues, and the reviewer has costed them at $855,000. Each section of this report states its own basis, so a reader can accept one section and reject another.

Grading in the present report is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. J. Ferreiro has left several concerns in the present report at limited evidence strength rather than dropping them or dressing them up. Interviewees for the present report were told that a stated uncertainty would be recorded as an uncertainty rather than resolved against them. 7 teams contributed to the interview programme for the report at TDR-2025-17, and one of them was interviewed twice after a document arrived late. Figures quoted in the present report come from the company's systems unless the text says the reviewer measured them.

Report TDR-2025-17 is limited by log retention: the telemetry window is ninety days, so events before 2026-02-20 could not be examined directly. TDR-2025-17 carries one structural limitation: the company's own metrics are the only source for several objectives, and the reviewer could not build an independent measurement inside the engagement. The report at TDR-2025-17 distinguishes three positions: verified defect, credible concern, and matter requiring further work. The third category is not empty. This report reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. Calder Rowe Engineering Advisory has reported unresolved disagreements as disagreements rather than choosing between the accounts.

17 interviews inform the present report. Where an interview is the sole source for a claim, the claim is graded limited. Minutes of every this report interview were returned to the participant, and three came back with substantive corrections. J. Ferreiro would put 6 items from this report into the first hundred days rather than into a warranty. This report recommends that 1 items be reflected in the disclosure schedules and that the remainder be tracked in the post-close plan. J. Ferreiro has recorded the date of each measurement, because several of them will be stale within a quarter.

Schedule J - Site deployments, capacity headroom and integration target (TDR-2025-17)

Site	Customer	Robots deployed	Go-live	Capacity headroom	Warehouse system	Note
Fremont, California	internal engineering site	34	2021-06	not measured	test harness	the reference site, and the only one with a laboratory alongside
Meridian, Idaho	Asterion Logistics	96	2022-09	22%	vendor A	the longest-running customer

Site	Customer	Robots deployed	Go-live	Capacity headroom	Warehouse system	Note
						deployment and the source of the most incident history
Grand Prairie, Texas	Asterion Logistics	78	2023-04	31%	vendor A	the second Asterion site, commissioned on the same template
Wilmington, Ohio	Crestline Fulfillment	112	2023-08	14%	vendor B	the largest deployment and the closest to the allocator scaling limit
Lehigh Valley, Pennsylvania	Crestline Fulfillment	64	2024-02	38%	vendor B	commissioned during the connector rework
Ontario, California	Evergreen Warehousing	88	2024-05	26%	vendor C	the site whose shift pattern change caused the April 2025 charge incident
Groveport, Ohio	Granite Peak Distribution	52	2024-09	44%	vendor A	opportunity charging only, which is where the pack estimator drift appeared
Kenosha, Wisconsin	Dunbar Industrial	41	2024-11	49%	vendor D	the second opportunity-charging site
Tracy, California	Harborline Systems	58	2025-01	36%	vendor B	the site whose cleaning schedule now overlaps the peak shift
Shepherdsville, Kentucky	Fathom Mobility	47	2025-03	41%	vendor C	manipulator-equipped robots are concentrated here
Reno, Nevada	Blue Mesa Components	29	2025-05	57%	vendor D	the smallest customer deployment
Newnan, Georgia	Ironwood Grocery Group	66	2025-07	33%	vendor B	narrow-aisle geometry, and the source of the August 2025 mutual-yield stall
Whitestown, Indiana	Juniper Bay Apparel	44	2025-09	45%	vendor C	the site with the standing map publication workaround
Bethlehem, Pennsylvania	Kestrel Medical Supply	37	2025-11	51%	vendor A	sparse survey features, which is why relocalisation is slow here
Chino, California	Larkspur Home	33	2026-01	54%	vendor D	the go-live delayed by the forked provisioning template
Edwardsville, Illinois	Crestline Fulfillment	24	2026-03	63%	vendor B	the newest Crestline site, still ramping
Lockbourne, Ohio	Granite Peak Distribution	19	2026-04	68%	vendor A	commissioned after the template baseline check landed
Mesquite, Texas	Evergreen Warehousing	16	2026-06	71%	vendor C	in acceptance testing at the review date

2. Load-test programme and the LT-2026-15 result

Measurement of 1000-robot run LT-2026-15 is against the fifteen-minute map publication objective rather than against a customer commitment. What run LT-2026-15 does and does not establish is worth stating: the test environment carried a smaller dataset than production, so the database term is understated here. Errors in LT-2026-15 were 0.10%, with latency of 3 m 20 s, 8 m 40 s and 13 m 10 s across the three percentiles. 1000-robot run LT-2026-15 is included because a reader of TDR-2025-17 should be able to disagree with the inference without repeating the work.

720 robots and 13,800 orders an hour define load run LT-2026-16, run on 2026-04-17. The percentile set for load run LT-2026-16 reads 1 m 50 s, 4 m 30 s and 9 m 20 s, with errors at 0.00%. Set beside the other runs, load run LT-2026-16 varies only in the size of the concurrent operator session pool. scenario LT-2026-16 illustrates why TDR-2025-17 separates the severity of a finding from the confidence behind it.

The test bed for run LT-2025-11 carries production data volumes and a stubbed paging provider, which is the main reason the reviewer does not treat it as production-equivalent. Single site, steady state run LT-2025-11 exercised the single site, steady state case with 120 robots and 1,850 orders an hour, on 2025-10-08. The percentile set for LT-2025-11 reads 210 ms, 480 ms and 760 ms, with errors at 0.02%. single site, steady state run LT-2025-11 is one of the entries where TDR-2025-17 relies on the company's own measurement rather than an independent one.

Run LT-2025-12 identified the binding constraint: allocator database write latency became the leading term. Outcome of single site, peak hour run LT-2025-12 was pass, and the reviewer read the raw result rather than the summary. Measurement of single site, peak hour run LT-2025-12 is against the five-minute configuration distribution objective rather than against a customer commitment. load run LT-2025-12 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Scenario LT-2025-13 was judged against the fifteen-minute map publication objective. Scenario LT-2025-13 is the single site, peak with a charger outage run, sized at 120 robots and 2,640 orders an hour. The test bed for 120-robot run LT-2025-13 carries the production cloud configuration and a laboratory network profile, which is the main reason the reviewer does not treat it as production-equivalent. LT-2025-13 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Run LT-2025-14 is not read as a headline number: the test environment carried a smaller dataset than production, so the database term is understated here. Scenario LT-2025-14 produced a 290 ms median with 820 ms and 1,510 ms tails and an 0.09% error rate. In scenario LT-2025-14 the bottleneck was no limit reached. LT-2025-14 illustrates why TDR-2025-17 separates the severity of a finding from the confidence behind it.

Configuration for scenario LT-2025-15 was 480 robots against 9,400 orders an hour in the four sites, shared region case. In load run LT-2025-15 the median was 355 ms and the tail reached 2,240 ms; errors ran at 0.18%. The limiting term in scenario LT-2025-15 was traffic manager Redis connection pool saturated at 84 per cent. run LT-2025-15 was cross-checked against the engineering record supplied for TDR-2025-17, and the two agree.

Fail against the 1,200 ms p95 target is what the test record gives for scenario LT-2025-16. LT-2025-16 was designed to change one variable, namely the number of sites in the region rather than the order rate. Latency in 720-robot run LT-2025-16 was 470 ms at p50, 1,680 ms at p95 and 3,410 ms at p99, with an error rate of 0.34%. scenario LT-2025-16 was reviewed against the position twelve months earlier, and nothing measurable has changed.

The result recorded for load run LT-2025-17 is fail against the 1,200 ms p95 target, which the reviewer traced to the underlying measurement. Set beside the other runs, order burst, ten times nominal for two minutes run LT-2025-17 varies only in the number of sites in the region rather than the order rate. Load run LT-2025-17 ran against an environment with a smaller order history than production and a synthetic warehouse endpoint. load run LT-2025-17 was cross-checked against the engineering record supplied for TDR-2025-17, and the two agree.

Schedule P - Load-test matrix against the published latency targets (TDR-2025-17)

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
LT-2025-11	single site, steady state	120	1,850	210 ms	480 ms	760 ms	0.02%	pass
LT-2025-12	single site, peak hour	120	2,640	265 ms	690 ms	1,240 ms	0.05%	pass

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
LT-2025-13	single site, peak with a charger outage	120	2,640	310 ms	870 ms	1,690 ms	0.11%	pass with observation
LT-2025-14	two sites, shared region	240	4,900	290 ms	820 ms	1,510 ms	0.09%	pass
LT-2025-15	four sites, shared region	480	9,400	355 ms	1,120 ms	2,240 ms	0.18%	pass with observation
LT-2025-16	six sites, shared region	720	13,800	470 ms	1,680 ms	3,410 ms	0.34%	fail against the 1,200 ms p95 target
LT-2025-17	order burst, ten times nominal for two minutes	240	12,000	620 ms	2,350 ms	5,900 ms	1.24%	fail against the 1,200 ms p95 target
LT-2025-18	telemetry burst with control traffic on the shared topic	480	9,400	540 ms	4,100 ms	11,300 ms	0.42%	fail
LT-2025-19	region failover during peak	480	9,400	410 ms	1,950 ms	6,700 ms	2.85%	fail against the error rate target
LT-2025-20	database failover, allocator primary	240	4,900	340 ms	1,410 ms	4,200 ms	0.96%	pass with observation
LT-2026-01	six sites after the topic split	720	13,800	395 ms	1,090 ms	2,050 ms	0.14%	pass
LT-2026-02	eight sites, shared region	960	18,200	455 ms	1,340 ms	2,880 ms	0.27%	fail against the 1,200 ms p95 target
LT-2026-03	eight sites with the allocator read replicas enabled	960	18,200	400 ms	1,150 ms	2,310 ms	0.21%	pass
LT-2026-04	ten sites, shared region	1200	22,600	520 ms	1,720 ms	4,050 ms	0.48%	fail against the 1,200 ms p95 target
LT-2026-05	order burst after the broker queue change	480	24,000	480 ms	1,240 ms	2,760 ms	0.31%	pass with observation
LT-2026-06	shard rebalance under peak load	720	13,800	430 ms	1,280 ms	3,120 ms	0.36%	pass with observation
LT-2026-07	identity service token storm	960	18,200	180 ms	520 ms	1,010 ms	0.03%	pass
LT-2026-08	console session concurrency	480	9,400	290 ms	740 ms	1,480 ms	0.07%	pass
LT-2026-09	console session concurrency at twice the plan	480	9,400	380 ms	1,460 ms	3,900 ms	0.52%	fail
LT-2026-10	warehouse connector burst, one tenant	240	8,800	610 ms	1,830 ms	4,400 ms	0.68%	fail against the 1,500 ms p95 acknowledgement target
LT-2026-11	warehouse connector burst after outbox partitioning	240	8,800	440 ms	1,210 ms	2,540 ms	0.29%	pass
LT-2026-12	telemetry	2400	n/a	1.4 s to	22 s	48 s	0.08%	pass

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
	ingest at twice the current fleet			console				
LT-2026-13	twelve sites, projected 2027 fleet	1440	27,000	690 ms	2,450 ms	6,100 ms	0.74%	fail
LT-2026-14	four hour edge buffer drain after an outage	720	13,800	not applicable	not applicable	not applicable	0.00%	pass with observation
LT-2026-15	map publication to a thousand robots	1000	n/a	3 m 20 s	8 m 40 s	13 m 10 s	0.10%	pass
LT-2026-16	config distribution with thirty per cent of robots offline	720	13,800	1 m 50 s	4 m 30 s	9 m 20 s	0.00%	pass with observation

3. Latency percentiles against the published targets

480-robot run LT-2025-18 ran against an environment with production-sized data but a single availability zone. Errors in run LT-2025-18 were 0.42%, with latency of 540 ms, 4,100 ms and 11,300 ms across the three percentiles. What load run LT-2025-18 does and does not establish is worth stating: one run does not establish a trend, and this scenario was executed once. telemetry burst with control traffic on the shared topic run LT-2025-18 appears twice in the working papers for TDR-2025-17 because the first answer was corrected after a follow-up request.

Run LT-2025-19 was executed in the performance environment, which carries the current production topology with one region held out. Run LT-2025-19 exercised the region failover during peak case with 480 robots and 9,400 orders an hour, on 2025-10-23. Run LT-2025-19 sits against the fifteen-minute map publication objective; whether that target is the right one is a separate question. LT-2025-19 is included because a reader of TDR-2025-17 should be able to disagree with the inference without repeating the work.

1,410 ms at p95 is the figure that matters in load run LT-2025-20, against 340 ms median and 4,200 ms at the tail. On run LT-2025-20 the reviewer notes that the headroom claim is supportable at the tested scale and not at the scale in the operating plan. Applying the 1,500 ms p95 acknowledgement objective for the warehouse connector to LT-2025-20 produces the verdict recorded above. database failover, allocator primary run LT-2025-20 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The environment behind run LT-2026-01 carries the production cloud configuration and a laboratory network profile, and the reviewer weighted the result accordingly. Applying the five-minute configuration distribution objective to six sites after the topic split run LT-2026-01 produces the verdict recorded above. Set beside the other runs, scenario LT-2026-01 varies only in the number of sites in the region rather than the order rate. load run LT-2026-01 is the reason one section of TDR-2025-17 concludes narrowly rather than generally.

960-robot run LT-2026-02 is the eight sites, shared region run, sized at 960 robots and 18,200 orders an hour. The environment behind eight sites, shared region run LT-2026-02 carries production data volumes and a stubbed paging provider, and the reviewer weighted the result accordingly. The target scenario LT-2026-02 was measured against is the five-minute configuration distribution objective. load run LT-2026-02 is one of the entries where TDR-2025-17 relies on the company's own measurement rather than an independent one.

Measured latency in run LT-2026-03 was 400 ms at the median, 1,150 ms at p95 and 2,310 ms at p99. Load run LT-2026-03 exercised the eight sites with the allocator read replicas enabled case with 960 robots and 18,200 orders an hour, on 2026-02-13. The limiting term in run LT-2026-03 was read offload bought headroom without a schema change. scenario LT-2026-03 is treated as material to the investment case rather than as a routine engineering item.

On run LT-2026-04 the reviewer notes that the run was executed against a synthetic warehouse system rather than a customer endpoint, which flatters the tail. Scenario LT-2026-04 ran against an environment with production-sized data but a single availability zone. The applicable target for 1200-robot run LT-2026-04 is the 1,200 ms p95 objective

published for order assignment, and the reviewer checked where that target came from. ten sites, shared region run LT-2026-04 is reported here in full because a summary of it in an earlier draft of TDR-2025-17 was misread as an assurance.

The test bed for scenario LT-2026-05 carries the current production topology with one region held out, which is the main reason the reviewer does not treat it as production-equivalent. Order burst after the broker queue change run LT-2026-05 is scored against the 0.5 per cent error-rate ceiling in the internal service documentation, which the company set in its own service documentation. Outcome of order burst after the broker queue change run LT-2026-05 was pass with observation, and the reviewer read the raw result rather than the summary. load run LT-2026-05 is reported at the evidence strength available and has not been promoted to fit the narrative.

On LT-2026-06 the reviewer notes that the error rate rather than the latency is the term that would first breach a customer commitment. The applicable target for run LT-2026-06 is the thirty-second telemetry visibility objective, and the reviewer checked where that target came from. Set beside the other runs, 720-robot run LT-2026-06 varies only in the order rate rather than the fleet size. load run LT-2026-06 would move from a concern to a verified defect if one further measurement were supplied.

Schedule H - Change volume, change failure rate and restore time by month (TDR-2025-17)

Month	Changes to production	Failed changes	Change failure rate	Mean time to restore	Rollbacks
2025-01	70	0	0.0%	57 min	0
2025-02	67	3	4.4%	59 min	3
2025-03	40	3	7.5%	47 min	1
2025-04	74	0	0.0%	66 min	0
2025-05	41	1	2.4%	41 min	2
2025-06	63	4	6.3%	22 min	1
2025-07	64	2	3.1%	62 min	3
2025-08	73	0	0.0%	41 min	0
2025-09	80	7	8.7%	24 min	0
2025-10	77	3	3.8%	38 min	0
2025-11	61	1	1.6%	84 min	2
2025-12	62	4	6.4%	21 min	3
2026-01	71	5	7.0%	59 min	2
2026-02	46	3	6.5%	20 min	2
2026-03	49	1	2.0%	50 min	1
2026-04	61	4	6.5%	33 min	2
2026-05	51	1	1.9%	20 min	0
2026-06	81	1	1.2%	42 min	0
2026-07	78	1	1.2%	86 min	2
2026-08	39	3	7.6%	26 min	3

4. The services that bound throughput

Speed and Separation Monitor rests on the separation monitoring design note, the internal safety case and six bench approach tests so far as this review is concerned. Speed and Separation Monitor was built in C++17 with a safety-reviewed core and has not been rewritten; it still runs on on-robot Linux, pinned to the safety compute path against a rolling in-memory window with fault snapshots to the safety controller. On Speed and Separation Monitor the conclusion is qualified: the component is adequate today and carries a supportability liability that is not on the funded plan. The open defect backlog for Speed and Separation Monitor holds 4 items, the most consequential being that the protective distance calculation uses a fixed operator approach speed rather than a measured one. Speed and Separation Monitor sits edge, cross-checked against the safety controller's own inputs, and no alternative placement is currently supported. Speed and Separation Monitor illustrates why TDR-2025-17 separates the severity of a finding from the confidence behind it.

On paper Speed and Separation Monitor releases twice a year, with revalidation, and the observed practice did not diverge from that. Service-level intent for Speed and Separation Monitor states that commanded speed is reduced before the separation distance falls below the protective distance. Speed and Separation Monitor does not share a schema with any consumer; every read by the safety controller and the motion planner speed limit goes through the published interface. The ageing position for Speed and Separation Monitor is none; the sensing hardware is current production, and the reviewer checked it against the vendor statements on file. Reading the dependency map from Speed and Separation Monitor outward gives the costmap fusion pipeline and the safety-rated scanner inputs on the input side and the safety

controller and the motion planner speed limit on the output side. Speed and Separation Monitor is the reason one section of TDR-2025-17 concludes narrowly rather than generally.

The assessment of OTA Updater is not unqualified: the component would not block the transaction, and it would appear on any competent post-close engineering plan. What the reviewer had for OTA Updater was the update state machine specification, signing key custody notes and two rollback drills; the conclusions below do not reach past it. The Platform Infrastructure team carries OTA Updater on its service inventory, and Erik Palmgren answered for it in interview. OTA Updater carries 8 open defects, among them a case where a paused ring resumes automatically after a service restart, which surprised two site teams. The cadence recorded for OTA Updater is every three weeks, following the autonomy release, which matches the observed release history. OTA Updater is reported at the evidence strength available and has not been promoted to fit the narrative.

In production OTA Updater is split edge and cloud, with staged rings and a per-site pause control, unchanged since the last architecture review. Data ownership is clear for OTA Updater: it writes content-addressed object storage for artefacts, PostgreSQL 14 for state and no other service writes there. OTA Updater's dependency ageing was checked: none; artefact signing was moved to the current key format in 2025. The published objective for OTA Updater is that an update is either fully applied or fully rolled back, with no mixed-version robot. OTA Updater sits between the release orchestrator and the artefact signing service and the on-robot supervisor and the fleet health monitor, and that position is what makes its availability objective load-bearing. OTA Updater was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Underlying Fleet Allocator in this report are the allocator service contract, two design reviews and a week of assignment audit logs. Fleet Allocator is deployed cloud, one regional deployment serving all sites in that region. Fleet Allocator pairs a Go 1.22 implementation with cloud, containers on a managed Kubernetes cluster, and its data lives in PostgreSQL 14 with a synchronous standby. Read against its objective, Fleet Allocator the component is adequate today and carries a supportability liability that is not on the funded plan. Against Fleet Allocator, the coverage figure is 82 per cent line coverage, which the figure is respectable but excludes the fallback path, which is where the known defect sits. Fleet Allocator sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

9 defects remain open against Fleet Allocator; the one the reviewer pressed on is that assignment starvation for robots parked at chargers below the battery reserve threshold. Fleet Allocator is owned by the Fleet Services team, with Marta Kellerman named as the engineering owner of record. Fleet Allocator depends on the order broker and the robot shadow store and is depended on by the traffic manager, the charge scheduler and the fleet health monitor. Service-level intent for Fleet Allocator states that task assignment inside 900 ms of order arrival at p95. For Fleet Allocator the supportability position is none identified in the reviewed manifests. Fleet Allocator was reviewed against the position twelve months earlier, and nothing measurable has changed.

The Robot Firmware team carries Battery Management Service on its service inventory, and Peter Halvorsen answered for it in interview. Changes to Battery Management Service reach production pack firmware twice a year, the cloud service monthly. Behind Battery Management Service sits pack non-volatile memory and a cloud time-series store; the service itself is C on the pack controller, Python 3.11 in the cloud on split: pack firmware and a cloud analytics service. The charge scheduler and the fleet health monitor rely on Battery Management Service, which in turn relies on the pack sensors and the charge scheduler. The published objective for Battery Management Service is that pack cell imbalance is reported before it reaches the derating threshold. Battery Management Service is the reason one section of TDR-2025-17 concludes narrowly rather than generally.

Evidence examined for Battery Management Service comprised the pack firmware release notes, cell test data for two revisions and a derating incident file. Battery Management Service was examined for unsupported components; one pack revision uses a cell supplier that has since discontinued the part. Data ownership is clear for Battery Management Service: it writes pack non-volatile memory and a cloud time-series store and no other service writes there. 10 items sit in the Battery Management Service backlog; the entry that bears on the investment case is that state-of-health estimates drift on packs that spend most of their life on opportunity charge. Battery Management Service runs split edge and cloud; that placement is what determines whether a fault is one robot or one region. Battery Management Service illustrates why TDR-2025-17 separates the severity of a finding from the confidence behind it.

Config Distribution was assessed against the objective and the defect record, and the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. Rust 1.79 agent with a Go 1.22 control service is the implementation language for Config Distribution, which is hosted on split: an on-robot agent and a cloud service with content-addressed object storage with a PostgreSQL 14 index behind it. Config Distribution is promoted weekly, with the ring order fixed by policy. In production Config Distribution is split edge and cloud, with a signed bundle format, unchanged since the last architecture review. The backlog for Config Distribution is 6 items deep and contains an acknowledged defect whereby a robot that is offline at publication applies the change without re-validating preconditions. Config Distribution was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Config Distribution reports 72 per cent line coverage, measured by the same instrumentation the pipeline uses. Data ownership is clear for Config Distribution: it writes content-addressed object storage with a PostgreSQL 14 index and no other service writes there. Config Distribution has a single named owner, Sam Oyelaran of Platform Infrastructure, which is not true of every service reviewed. The objective recorded against Config Distribution is that a configuration change reaches every online robot inside five minutes, and it is reported weekly to the engineering leadership. The dependency map places the site provisioning service and the artefact signing service above Config Distribution and the on-robot supervisor and the motion planner parameters below it. Config Distribution is one of the entries where TDR-2025-17 relies on the company's own measurement rather than an independent one.

Service-level intent for Site Provisioning Service states that a new site reaches a working configuration inside one working day. Responsibility for Site Provisioning Service was confirmed with Marcus Duplessis, whose team, Platform Infrastructure, also holds the on-call duty. The cadence recorded for Site Provisioning Service is fortnightly, which matches the observed release history. Site Provisioning Service is written in Go 1.22, executes on cloud, containers on a managed Kubernetes cluster, and keeps its state in PostgreSQL 14 with a configuration history table. 5 items sit in the Site Provisioning Service backlog; the entry that bears on the investment case is that site templates diverged after two customers required bespoke network settings. Site Provisioning Service was tested rather than accepted, and the test is described in the working papers behind TDR-2025-17.

Site Provisioning Service was tested against the dependency ageing register, which records none identified. Site Provisioning Service was weighed on its coverage, its defect backlog and its dependency position, and it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Inputs to Site Provisioning Service come from the identity service and the customer network intake records, and its consumers are the config distribution service and the WMS connector. Site Provisioning Service shows 64 per cent line coverage in the pipeline, and coverage alone overstates the assurance here, because the bench suite is the real safety net. For Site Provisioning Service the reviewer worked from the provisioning runbook, template diffs for four sites and two go-live records. Site Provisioning Service was checked against the corresponding entry in the company's own register, and the register was current.

Service-level intent for Safety Controller states that a validated protective-stop trigger de-energises the drive inside 120 ms. Safety Engineering releases Safety Controller twice a year, with full revalidation of the safety case, and no release in the sampled window bypassed the gate set. For Safety Controller the reviewer worked from the supplier assessment report, the internal safety case and eight bench trigger tests. Taking the evidence together, Safety Controller the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. Bettina Vogel leads Safety Engineering's work on Safety Controller and was the reviewer's principal interview for it. Safety Controller was raised again with the owning engineer after the initial interview, and the position did not change.

Supportability of Safety Controller turns on one fact: none; the supplier's support commitment runs to 2031. Coverage for Safety Controller was 91 per cent line coverage at the review date, and the trend over the reviewed period was down by two points after a large refactor. Reading the dependency map from Safety Controller outward gives the E-stop chain, the speed and separation monitor and the drive current sensors on the input side and the drive power contactors and the robot fault annunciator on the output side. Safety Controller owns its own data and exposes it only through a versioned interface, which the reviewer read. Of the 3 open items against Safety Controller, the reviewer treats one as material: the fault log wraps after four thousand entries and the oldest records are discarded. Safety Controller was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Changes to Analytics Warehouse Loader reach production weekly. The interface Analytics Warehouse Loader publishes is versioned, and the customer reporting views and the internal metrics layer were checked for use of any undocumented field. Automated test coverage on Analytics Warehouse Loader stands at 45 per cent line coverage. Upstream of Analytics Warehouse Loader sit the telemetry ingest path, the order broker and the billing usage meter; downstream, the customer reporting views and the internal metrics layer consume its output. The assessment of Analytics Warehouse Loader is not unqualified: the reviewer could corroborate the availability claim but not the latency claim, which rests on a single dashboard. Analytics Warehouse Loader was reviewed against the position twelve months earlier, and the direction of travel is favourable.

For Analytics Warehouse Loader the supportability position is one transformation still runs on a deprecated warehouse function. The evidence base for Analytics Warehouse Loader was the load schedule, partition completeness checks and two months of load audit records. Analytics Warehouse Loader is deployed cloud, one loader per region writing to a shared warehouse. Analytics Warehouse Loader was built in Python 3.11 with SQL transformations and has not been rewritten; it still runs on cloud, a scheduled batch runner against a columnar warehouse with daily partitions. Where Analytics Warehouse Loader is concerned, the commitment is that the daily load completes before 06:00 local time at every region; the reviewer traced the metric to its source. Analytics Warehouse Loader was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Schedule D - Automated coverage trend on the reviewed services (TDR-2025-17)

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Motion Planner	63%	70%	71%	+8	Autonomy Core
Costmap Fusion	51%	55%	58%	+7	Perception
Fleet Allocator	74%	79%	82%	+8	Fleet Services
Traffic Manager	65%	72%	74%	+9	Fleet Services
Charge Scheduler	54%	60%	63%	+9	Fleet Services
Safety Controller	82%	88%	91%	+9	Safety Engineering
E-Stop Chain	n/a	n/a	witnessed test	not comparable	Safety Engineering
Lidar Driver	37%	44%	44%	+7	Robot Firmware
Localization Engine	61%	66%	66%	+5	Autonomy Core
WMS Connector	60%	68%	69%	+9	Integrations
Order Broker	72%	75%	78%	+6	Fleet Services
Telemetry Ingest	55%	58%	61%	+6	Data and Telemetry
OTA Updater	67%	73%	76%	+9	Platform Infrastructure
Identity and Authorization	77%	82%	84%	+7	Security Engineering
Remote Operations Console	48%	51%	52%	+4	Applications
Simulation Harness	44%	46%	47%	+3	Developer Productivity
Hardware-in-the-Loop Rig	35%	36%	39%	+4	Robot Firmware

4. The services that bound throughput - part 2 of 2

Following on directly, these further rows of the 4. The services that bound throughput belong to the same schedule and share its numbering. The parts are to be read together.

The measurement basis recorded when the 4. The services that bound throughput opened is unchanged, so any figure here is comparable with any figure in the earlier parts. Nothing is restated from the earlier part.

Schedule D - Automated coverage trend on the reviewed services (TDR-2025-17) (continued, part 2 of 2)

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Fleet Health Monitor	50%	54%	57%	+7	Field Reliability
Battery Management Service	45%	51%	54%	+9	Robot Firmware
Gripper Controller	38%	38%	41%	+3	Robot Firmware
Map Authoring Studio	43%	48%	49%	+6	Applications
Site Provisioning Service	55%	62%	64%	+9	Platform Infrastructure
Event Bus	n/a	n/a	witnessed test	not comparable	Platform Infrastructure

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Config Distribution	68%	69%	72%	+4	Platform Infrastructure
Robot Shadow Store	63%	68%	68%	+5	Fleet Services
Analytics Warehouse Loader	42%	43%	45%	+3	Data and Telemetry
Billing Usage Meter	75%	77%	79%	+4	Data and Telemetry
Alarm Router	46%	50%	51%	+5	Field Reliability
Video Snippet Service	37%	43%	43%	+6	Applications
Speed and Separation Monitor	82%	87%	87%	+5	Safety Engineering
Wheel Odometry Filter	53%	60%	62%	+9	Robot Firmware
Payload Scale Interface	29%	33%	36%	+7	Robot Firmware
Edge Gateway Agent	67%	73%	74%	+7	Platform Infrastructure
Release Orchestrator	47%	53%	55%	+8	Developer Productivity

5. Capacity headroom site by site

The Reno, Nevada site integrates with vendor D and shows 57% headroom. At Reno, Nevada the fleet is 29 robots, and the site has been live since 2025-05. Reno, Nevada has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Headroom at Ontario, California stands at 26%; vendor C is the integrated warehouse system. Ontario, California is the site whose shift pattern change caused the April 2025 charge incident, and the reviewer weighted the sample accordingly. Ontario, California is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The Meridian, Idaho site integrates with vendor A and shows 22% headroom. The Meridian, Idaho deployment serves Asterion Logistics with 96 robots, live since 2022-09. Meridian, Idaho was reviewed against the position twelve months earlier, and nothing measurable has changed.

Internal engineering site operates 34 Helioscope robots at Fremont, California, commissioned in 2021-06. Test harness is the warehouse system at Fremont, California, where headroom is not measured. Fremont, California was excluded from an earlier draft of TDR-2025-17 and reinstated after the reviewer obtained the supporting record.

Against its peak profile, Grand Prairie, Texas retains 31% headroom on a vendor A integration. Grand Prairie, Texas is the second Asterion site, commissioned on the same template, and the reviewer weighted the sample accordingly. Grand Prairie, Texas was inspected in the repository rather than described from the ticket, which is why the account is specific.

Bethlehem, Pennsylvania is sparse survey features, which is why relocalisation is slow here, which is why the reviewer sampled it. Site ST-14 is Bethlehem, Pennsylvania: Kestrel Medical Supply, 37 robots, live from 2025-11. Bethlehem, Pennsylvania was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Headroom at Edwardsville, Illinois stands at 63%; vendor B is the integrated warehouse system. Edwardsville, Illinois came online in 2026-03 and now carries 24 robots for Crestline Fulfillment. Edwardsville, Illinois was one of two items where TDR-2025-17 could not obtain the underlying data within the engagement.

Crestline Fulfillment operates 64 Helioscope robots at Lehigh Valley, Pennsylvania, commissioned in 2024-02. For Lehigh Valley, Pennsylvania the material fact is that it is commissioned during the connector rework. Lehigh Valley, Pennsylvania is reported here in full because a summary of it in an earlier draft of TDR-2025-17 was misread as an assurance.

The Chino, California site integrates with vendor D and shows 54% headroom. What distinguishes Chino, California is that it is the go-live delayed by the forked provisioning template. Chino, California was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Remaining capacity at Kenosha, Wisconsin is 49%, measured against the site's peak order profile. The reason Kenosha, Wisconsin appears here is that it is the second opportunity-charging site. Kenosha, Wisconsin is treated as material to the investment case rather than as a routine engineering item.

For Whitestown, Indiana the material fact is that it is the site with the standing map publication workaround. Vendor C is the warehouse system at Whitestown, Indiana, where headroom is 45%. Whitestown, Indiana was reviewed against the position twelve months earlier, and the direction of travel is favourable.

2024-09 is the go-live month at Groveport, Ohio, where Granite Peak Distribution now runs 52 robots. Headroom at Groveport, Ohio stands at 44%; vendor A is the integrated warehouse system. Groveport, Ohio is the clearest instance in TDR-2025-17 of a control that exists on paper and is weaker in practice.

58 robots are deployed at Tracy, California, which went live in 2025-01. Vendor B is the warehouse system at Tracy, California, where headroom is 36%. Tracy, California is included because a reader of TDR-2025-17 should be able to disagree with the inference without repeating the work.

Newnan, Georgia is notable in this review because it is narrow-aisle geometry, and the source of the August 2025 mutual-yield stall. Remaining capacity at Newnan, Georgia is 33%, measured against the site's peak order profile. Newnan, Georgia carries a follow-up request that was satisfied after the draft of TDR-2025-17 was circulated.

On Mesquite, Texas the record shows that it is in acceptance testing at the review date. 2026-06 is the go-live month at Mesquite, Texas, where Evergreen Warehousing now runs 16 robots. Mesquite, Texas would move from a concern to a verified defect if one further measurement were supplied.

Capacity headroom at Wilmington, Ohio is 14%, and the warehouse system is vendor B. Wilmington, Ohio is the largest deployment and the closest to the allocator scaling limit, which is why the reviewer sampled it. Wilmington, Ohio sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Lockbourne, Ohio is notable in this review because it is commissioned after the template baseline check landed. Against its peak profile, Lockbourne, Ohio retains 68% headroom on a vendor A integration. Lockbourne, Ohio is the entry that changed most between the first and second document requests behind TDR-2025-17.

6. Capacity incidents and what they revealed

Following Crestline Fulfillment incident INC-2026-039 the team undertook to serialise cancellation against assignment and add an hourly orphan sweep, with Grace Lindqvist accountable by 2026-05-31. The reviewer grades the INC-2026-039 evidence as strong on the cause and limited on the contributing factor and has not upgraded the finding to compensate. The classification applied to concurrency incident INC-2026-039 is concurrency, and the affected component is Order Broker. Crestline Fulfillment incident INC-2026-039 closure status is landed on 2026-04-26. Taken with the rest of the record, Order Broker incident INC-2026-039 indicates that the incident is unremarkable in itself and revealing about the observability of the surrounding system. concurrency incident INC-2026-039 carries a follow-up request that was satisfied after the draft of TDR-2025-17 was circulated.

Concurrency event INC-2026-039 took 1 h 04 m to stabilise and 19 days to close out. Beyond the direct cause, INC-2026-039 shows that the journal reconciliation runs nightly, so the orphan was invisible until the morning. Underneath concurrency event INC-2026-039 lies the design decision that cancellation and assignment wrote to the task record without a shared lock. Concurrency incident INC-2026-039 began on 2026-04-07 when an order cancellation raced with assignment and left an orphaned task that blocked a pick face for an hour. Scope of impact in concurrency incident INC-2026-039 was one site and 1 robots, taken from the fleet records rather than from the narrative. Crestline Fulfillment incident INC-2026-039 is the entry that changed most between the first and second document requests behind TDR-2025-17.

In resource leak incident INC-2025-041 the first signal was a site supervisor telephoned the on-call engineer after robots queued at one intersection, and the reviewer treats that as reported detection. For resource leak event INC-2025-041 the reviewer holds moderate on the cause and strong on the delivery position, which was checked in the repository evidence and reports the finding at that strength. The contributing factor recorded against INC-2025-041 is that the alert rule for queue depth existed but was scoped to a metric the site did not emit. Mitigation of INC-2025-041 took 1 h 52 m and full resolution took 9 days. Remediation for Crestline Fulfillment incident INC-2025-041 is to give every corridor

reservation a lease with an expiry and add a site-level queue depth alert; Owen Brantley owns it and the committed date is 2025-05-30. Crestline Fulfillment incident INC-2025-041 was checked against the company's own register, which had not been updated since the change landed.

Cause analysis in INC-2025-041 concluded that the reservation lease had no expiry, so release depended on the departing robot. The blast radius of Traffic Manager incident INC-2025-041 was 2 sites and 61 robots. Taken with the rest of the record, incident INC-2025-041 indicates that the team's analysis was honest about a gap it could have concealed. The reviewer checked the Traffic Manager incident INC-2025-041 commitment against the code and release records; landed on 2025-05-22, verified in the reservation replay suite. Crestline Fulfillment incident INC-2025-041 records that on 2025-03-14, a robot lost its network link mid-traverse and its corridor reservation was never released. resource leak incident INC-2025-041 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Mitigation of Harborline Systems incident INC-2025-078 took 1 h 15 m and full resolution took 34 days. On 2025-06-24 the record for incident INC-2025-078 opens: the console listed robots that had been removed from a site and an operator commanded a decommissioned unit. INC-2025-078 was caused by a design gap: the session list cached the site roster and only invalidated it on a page reload. As to whether the incident INC-2025-078 remediation actually landed: landed on 2025-07-28; the regression test was still marked flaky at review. So far as reach is concerned, Harborline Systems incident INC-2025-078 involved 4 site or tenant with no robot taken out of service. incident INC-2025-078 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

How stale state incident INC-2025-078 surfaced matters to the assessment: an internal engineer noticed the discrepancy during a demonstration. Stale state is the class of Harborline Systems incident INC-2025-078; Remote Operations Console is the component. Harborline Systems incident INC-2025-078 would have been shorter or smaller had it not been that no test covered a roster change during an open operator session. What stale state event INC-2025-078 tells the reviewer is that the failure was in the detection design rather than in the code that failed. After Harborline Systems incident INC-2025-078, Anjali Sekhon took the action to invalidate the roster cache on a fleet membership event and add a session-level regression test by 2025-08-31. INC-2025-078 was checked against the company's own register, which had not been updated since the change landed.

Hardware fault incident INC-2025-090 traces back to a single fact: a marginal connector on one channel produced intermittent readings inside the accepted noise band. For incident INC-2025-090 the reviewer holds strong, since the reviewer reproduced the failure mode in a laboratory environment evidence and reports the finding at that strength. The reviewer treats one contributing factor in incident INC-2025-090 as the more important lesson: the connector torque specification was not part of the incoming inspection sheet. Verification of the INC-2025-090 remediation gives the inspection change landed on 2025-08-04; nine units await retrofit. The blast radius of Fathom Mobility incident INC-2025-090 was one site and 3 robots. hardware fault event INC-2025-090 is reported here in full because a summary of it in an earlier draft of TDR-2025-17 was misread as an assurance.

INC-2025-090 supports one conclusion and not another: the blast radius was set by a deployment topology decision made years earlier. Following incident INC-2025-090 the team undertook to add the connector torque check to incoming inspection and retrofit the affected harness batch, with Bettina Vogel accountable by 2025-09-30. Fathom Mobility incident INC-2025-090 was raised on 2025-07-19 after a protective stop was triggered by a false positive on one safety channel while the second channel read clear. How incident INC-2025-090 surfaced matters to the assessment: the robot self-reported the fault and the safety case review picked it up. Fathom Mobility incident INC-2025-090 was mitigated in immediate stop, 22 m to return to service; the resolution ran to 41 days. INC-2025-090 is the clearest instance in TDR-2025-17 of a control that exists on paper and is weaker in practice.

The corrective action for process defect incident INC-2025-146 is to fold the reset into the re-run command and make double-counting detectable in the ledger, and 2025-12-31 is the date the team gave. The two durations recorded for process defect incident INC-2025-146 are not applicable; caught before billing to mitigate and 11 days to resolve. Incident INC-2025-146 classifies as a process defect event against Billing Usage Meter. What went wrong in process defect event INC-2025-146 was that the re-run procedure required a manual ledger reset that the operator skipped. For Billing Usage Meter incident INC-2025-146 the reviewer holds strong, resting on the production traces the team supplied

evidence and reports the finding at that strength. process defect event INC-2025-146 illustrates why TDR-2025-17 separates the severity of a finding from the confidence behind it.

How Larkspur Home incident INC-2025-146 surfaced matters to the assessment: the monthly reconciliation caught it before invoicing. The contributing factor recorded against incident INC-2025-146 is that the runbook described the reset in prose rather than making it part of the command. The blast radius of Larkspur Home incident INC-2025-146 was no production site and no robot directly, the exposure being to internal systems. The trigger recorded for Larkspur Home incident INC-2025-146 on 2025-11-19 was that a re-run of a failed metering window double-counted usage for one customer. For process defect incident INC-2025-146 the follow-through was tested rather than assumed, and landed on 2025-11-30. process defect event INC-2025-146 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

The trigger recorded for Config Distribution incident INC-2026-015 on 2026-02-09 was that a robot that had been offline at publication applied a configuration bundle without re-validating preconditions. Fathom Mobility incident INC-2026-015 is not read as an isolated failure: the incident is unremarkable in itself and revealing about the observability of the surrounding system. Evidence strength on the INC-2026-015 analysis is limited, being a single account corroborated only by the ticket timeline. How validation event INC-2026-015 surfaced matters to the assessment: the on-robot supervisor raised a validation fault on the next mission. 44 m elapsed before Fathom Mobility incident INC-2026-015 was mitigated, with resolution at 22 days. validation incident INC-2026-015 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Cause analysis in Config Distribution incident INC-2026-015 concluded that precondition validation ran at publication time rather than at apply time. Contributing to validation incident INC-2026-015, and not repaired by the remediation, is that the offline apply path was added for a customer with intermittent connectivity and was lightly reviewed. After incident INC-2026-015, Sam Oyelaran took the action to move precondition validation to apply time and add an offline apply case to the platform suite by 2026-03-31. The classification applied to INC-2026-015 is validation, and the affected component is Config Distribution. On delivery, INC-2026-015 shows this: landed on 2026-03-03. Config Distribution incident INC-2026-015 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

The commitment register shows Asterion Logistics incident INC-2026-033 against an undertaking to measure closing rate where the sensing permits and re-run the protective distance calculation, with Bettina Vogel named and 2026-08-31 committed. The classification applied to incident INC-2026-033 is safety margin, and the affected component is Speed and Separation Monitor. Scope of impact in safety margin event INC-2026-033 was one site and 6 robots, taken from the fleet records rather than from the narrative. Root cause in INC-2026-033 was that the protective distance uses a fixed assumed approach speed rather than a measured closing rate. For INC-2026-033 the follow-through was tested rather than assumed, and not landed; a design study is running and the vendor assessment is booked. safety margin incident INC-2026-033 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Duration for Asterion Logistics incident INC-2026-033 splits into immediate speed reduction; 3 h to review the fleet of impact and open at the review date to close the underlying cause. The reviewer treats one contributing factor in INC-2026-033 as the more important lesson: the assumption was documented in the internal safety case and accepted at the time. Grading safety margin event INC-2026-033 on evidence rather than on consequence gives limited; the log retention window had closed before the reviewer asked for the raw data. INC-2026-033 supports one conclusion and not another: the fix is real and the regression protection is not, so recurrence remains possible. Attention came to Asterion Logistics incident INC-2026-033 when the on-robot safety log flagged a margin breach that did not become a contact event. INC-2026-033 was raised again with the owning engineer after the initial interview, and the position did not change.

Moderate, resting on the team's own post-mortem and one supporting log extract is the evidence grade for INC-2025-063, and the reader should not read severity into it. Attention came to INC-2025-063 when an operator noticed idle robots and raised a support ticket. Contributing to policy defect event INC-2025-063, and not repaired by the remediation, is that the charge scheduler and the allocator hold separate views of charger availability. Fleet Allocator incident INC-2025-063 is recorded as follows on delivery: landed on 2025-06-08. The two durations recorded for

INC-2025-063 are 6 h 05 m to mitigate and 12 days to resolve. INC-2025-063 was raised again with the owning engineer after the initial interview, and the position did not change.

Fleet Allocator owns policy defect incident INC-2025-063, which the reviewer classes as policy defect. INC-2025-063 is not read as an isolated failure: the team's analysis was honest about a gap it could have concealed. Granite Peak Distribution incident INC-2025-063 touched 3 sites and 28 robots, and Granite Peak Distribution was the affected account. Remediation for policy defect event INC-2025-063 is to make the allocator consult charger availability before excluding a robot; Marta Kellerman owns it and the committed date is 2025-07-31. Underneath INC-2025-063 lies the design decision that the allocator filter excluded low-charge robots without checking whether a charge slot was actually available. INC-2025-063 illustrates why TDR-2025-17 separates the severity of a finding from the confidence behind it.

Schedule F - Incident population examined, with follow-through status (TDR-2025-17)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-041	2025-03-14	Traffic Manager	2	61	1 h 52 m	9 days	resource leak	landed
INC-2025-047	2025-04-02	Lidar Driver	1	12	38 m	3 days	timing	partly landed
INC-2025-052	2025-04-21	Charge Scheduler	1	44	4 h 10 m	16 days	scheduling	landed
INC-2025-058	2025-05-09	WMS Connector	2	0	2 h 35 m	21 days	data integrity	landed
INC-2025-063	2025-05-27	Fleet Allocator	3	28	6 h 05 m	12 days	policy defect	landed
INC-2025-071	2025-06-11	Telemetry Ingest	6	214	52 m	52 m	availability	landed
INC-2025-078	2025-06-24	Remote Operations Console	4	0	1 h 15 m	34 days	stale state	landed
INC-2025-084	2025-07-08	OTA Updater	1	11	27 m	5 days	state persistence	landed
INC-2025-090	2025-07-19	Safety Controller	1	3	immediate stop, 22 m to return to service	41 days	hardware fault	partly landed
INC-2025-096	2025-08-05	Alarm Router	0	0	not applicable; no service impact	26 days	process gap	partly landed
INC-2025-103	2025-08-22	Motion Planner	1	2	19 m	58 days	algorithm defect	partly landed
INC-2025-110	2025-09-03	Analytics Warehouse Loader	1	0	not applicable; reporting only	18 days	data integrity	landed
INC-2025-118	2025-09-17	Identity and Authorization	0	0	3 h 40 m to revoke and rotate	7 days	secret exposure	partly landed
INC-2025-124	2025-10-01	Robot Shadow Store	5	178	1 h 08 m	29 days	consistency	landed
INC-2025-131	2025-10-14	Map Authoring Studio	1	34	3 h 22 m	44 days	process bypass	partly landed

6. Capacity incidents and what they revealed - part 2 of 2

Below, the 6. Capacity incidents and what they revealed continues; nothing separates this part from the last except the page on which it happens to fall. No row is omitted at the join.

The measurement basis recorded when the 6. Capacity incidents and what they revealed opened is unchanged, so any figure here is comparable with any figure in the earlier parts. No row is omitted at the join.

Schedule F - Incident population examined, with follow-through status (TDR-2025-17) (continued, part 2 of 2)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-	2025-11-02	Event Bus	7	246	2 h 47 m	63 days	capacity	partly landed

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
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INC-2025-146	2025-11-19	Billing Usage Meter	0	0	not applicable; caught before billing	11 days	process defect	landed
INC-2025-152	2025-12-04	Video Snippet Service	0	0	not applicable; retention only	37 days	data retention	partly landed
INC-2026-004	2026-01-13	Localization Engine	1	18	not a service outage; 5 days to a workaround	open at the review date	performance	not landed
INC-2026-009	2026-01-27	Gripper Controller	1	4	1 h 30 m to quarantine the affected robots	48 days	calibration	landed
INC-2026-015	2026-02-09	Config Distribution	2	7	44 m	22 days	validation	landed
INC-2026-021	2026-02-24	Simulation Harness	0	0	not applicable; developer tooling	open at the review date	tooling	not landed
INC-2026-028	2026-03-10	Fleet Health Monitor	1	22	not a service outage; 2 days of delayed repair	31 days	observability	landed
INC-2026-033	2026-03-25	Speed and Separation Monitor	1	6	immediate speed reduction; 3 h to review the fleet	open at the review date	safety margin	not landed
INC-2026-039	2026-04-07	Order Broker	1	1	1 h 04 m	19 days	concurrency	landed
INC-2026-045	2026-04-21	Edge Gateway Agent	1	39	5 h 12 m	26 days	observability	partly landed
INC-2026-051	2026-05-06	Site Provisioning Service	1	0	2 h 18 m	9 days	configuration drift	partly landed
INC-2026-057	2026-05-19	Battery Management Service	2	5	not a service outage; robots derated	open at the review date	modelling	not landed
INC-2026-062	2026-06-02	Release Orchestrator	0	0	not applicable; process only	open at the review date	process control	not landed
INC-2026-068	2026-06-16	Wheel Odometry Filter	1	14	not a service outage; 3 h of degraded operation	open at the review date	environmental	not landed

7. The scalability programme, its sequence and its cost

Exposure carried by register item TD-17 is that it a single key holder can sign firmware. Effort on security item TD-17 is estimated at 0.5 engineer-quarters, costed at $72,000. As at the review date security item TD-17 was in design, with 2026-11-30 still the stated target. TD-17 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Support item TD-03 cannot start earlier than it does because it is behind a platform change that has its own target date. Item TD-03 on Costmap Fusion in the debt register is to retire the vendored geometry library in costmap fusion, and it bears on Costmap Fusion. Effort on register item TD-03 is estimated at 1 engineer-quarter, costed at $128,000. register item TD-03 is treated as material to the investment case rather than as a routine engineering item.

The estimate for programme item TD-07 is 0.5 engineer-quarters and $68,000. Funded in the current financial year is the funding position on TD-07. Tooling item TD-07 cannot start earlier than it does because it is last in a three-item

chain, so its date is the least reliable on the register. programme item TD-07 appears twice in the working papers for TDR-2025-17 because the first answer was corrected after a follow-up request.

For programme item TD-12 the team estimates 1 engineer-quarter, which the reviewer converted to $144,000 for comparison. Resilience item TD-12 is the item to extend the edge buffer to twelve hours, classified under resilience. Reading the programme plan, resilience item TD-12 is behind a supplier release outside the company's control. resilience item TD-12 is the reason one section of TDR-2025-17 concludes narrowly rather than generally.

The estimate for programme item TD-16 is 1 engineer-quarter and $126,000. Security item TD-16 cannot start earlier than it does because it is after the determinism work in the simulation harness. 2026-10-15 is the target for TD-16, whose status is committed. Localization Engine item TD-16 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Localization Engine item TD-02 is sequenced in parallel with the release-process changes. The plan treats programme item TD-02 as funded by reallocating effort from feature work, which the reviewer checked against the headcount model. The reviewer treats this as both an IP and a supportability item is the point the reviewer would put to management on register item TD-02. register item TD-02 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Reading security item TD-15 against the funded plan, raised in the 2026 assessment and accepted for remediation. Progress on programme item TD-15 is recorded as committed against a 2026-09-30 date. $34,000 is the costed estimate for TD-15 on Alarm Router, based on 0.25 engineer-quarters. register item TD-15 rests on a single source, and the reviewer has said so rather than implying corroboration.

Under support, entry TD-09 is to upgrade the streaming platform before its support window closes. Register item TD-09 carries an estimate of 1.5 engineer-quarters; the costed figure is $196,000. Progress on TD-09 on Event Bus is recorded as committed against a 2027-06-30 date. programme item TD-09 was cross-checked against the engineering record supplied for TDR-2025-17, and the two agree.

Performance item TD-13 is sized at 1.5 engineer-quarters, or $178,000 at the company's loaded rate. 2026-12-31 is the target for TD-13 on Localization Engine, whose status is in design. If programme item TD-13 is not done, the exposure is that it relocalisation exceeds the operator prompt timeout at sparse sites. programme item TD-13 was cross-checked against the engineering record supplied for TDR-2025-17, and the two agree.

Hardware-in-the-Loop Rig item TD-05 carries an estimate of 0.5 engineer-quarters; the costed figure is $96,000. On funding, TD-05 is partly funded, with the second phase unbudgeted. As at the review date TD-05 on Hardware-in-the-Loop Rig was committed, with 2026-09-30 still the stated target. register item TD-05 would be a candidate for the first post-close engineering review under any reasonable plan.

Sequencing of TD-04 on Costmap Fusion matters: it is behind a customer maintenance window that is not yet agreed. Register item TD-04 is on the register because it defect escape rate on the fusion path is the highest in the fleet. Programme item TD-04 carries this observation: coverage moved from 51 to 58 per cent over the reviewed period. programme item TD-04 is the entry that changed most between the first and second document requests behind TDR-2025-17.

Raised by the reviewer rather than present on the company's own register is the point the reviewer would put to management on programme item TD-23. The estimate for Charge Scheduler item TD-23 is 0.75 engineer-quarters and $92,000. Status on TD-23 on Charge Scheduler is committed; the target is 2026-12-31. reliability item TD-23 illustrates why TDR-2025-17 separates the severity of a finding from the confidence behind it.

Effort on TD-10 is estimated at 0.5 engineer-quarters, costed at $54,000. Support item TD-10 is the item to migrate the deprecated warehouse transformation function, classified under support. As at the review date TD-10 was committed, with 2026-12-31 still the stated target. support item TD-10 is reported at the evidence strength available and has not been promoted to fit the narrative.

Register item TD-21 is partly funded, with the second phase unbudgeted; the register status and the funding position are not the same thing. Effort on TD-21 on Battery Management Service is estimated at 1 engineer-quarter,

costed at $138,000. In the programme quality item TD-21 is behind a platform change that has its own target date. TD-21 was raised again with the owning engineer after the initial interview, and the position did not change.

Register item TD-19 is behind a platform change that has its own target date, which is why the target date is later than the effort estimate implies. The reviewer's comment on TD-19 on Speed and Separation Monitor is that the most significant open safety item in the reviewed record. The register entry TD-19 on Speed and Separation Monitor reads: measure closing rate for the separation monitor, in the safety category. programme item TD-19 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Sequencing of safety item TD-20 matters: it is behind a platform change that has its own target date. Register item TD-20 is funded in the current financial year in the engineering plan the reviewer was shown. Status on safety item TD-20 is in progress; the target is 2026-09-15. Wheel Odometry Filter item TD-20 was checked against the company's own register, which had not been updated since the change landed.

TD-06 on Simulation Harness is blocked with a target date of 2026-11-30. The reviewer's comment on Simulation Harness item TD-06 is that blocked on the physics core determinism work. On funding, programme item TD-06 is not funded, and carried in the register regardless. register item TD-06 is the clearest instance in TDR-2025-17 of engineering practice being better than the documentation suggests.

The reviewer's qualification on Safety Controller item TD-18 is that the safety team supports it; it is not funded in the current plan. For Safety Controller item TD-18 the team estimates 0.75 engineer-quarters, which the reviewer converted to $104,000 for comparison. TD-18 is funded only if the team's headcount plan is met in the engineering plan the reviewer was shown. Safety Controller item TD-18 is reported at the evidence strength available and has not been promoted to fit the narrative.

8. Resilience runs: failover, rebalance and buffer drain

Scenario LT-2026-07 supports this conclusion: the reviewer would repeat this run with the newest site's order profile before relying on it. The percentile set for scenario LT-2026-07 reads 180 ms, 520 ms and 1,010 ms, with errors at 0.03%. 960-robot run LT-2026-07 was designed to change one variable, namely whether control and telemetry shared a topic. identity service token storm run LT-2026-07 was excluded from an earlier draft of TDR-2025-17 and reinstated after the reviewer obtained the supporting record.

In the series 480-robot run LT-2026-08 is the run that changes the size of the concurrent operator session pool. Load run LT-2026-08, taken with the other runs in the series, shows that the improvement between the two runs is real and its durability is untested under a shard rebalance. In load run LT-2026-08 the median was 290 ms and the tail reached 1,480 ms; errors ran at 0.07%. LT-2026-08 illustrates why TDR-2025-17 separates the severity of a finding from the confidence behind it.

The target LT-2026-09 was measured against is the 1,500 ms p95 acknowledgement objective for the warehouse connector. Set beside the other runs, scenario LT-2026-09 varies only in the warehouse system endpoint used. 480-robot run LT-2026-09 is not read as a headline number: the run was executed against a synthetic warehouse system rather than a customer endpoint, which flatters the tail. run LT-2026-09 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Run LT-2026-10 exercised the warehouse connector burst, one tenant case with 240 robots and 8,800 orders an hour, on 2026-02-26. Production-sized data but a single availability zone describes the environment warehouse connector burst, one tenant run LT-2026-10 ran in. Load run LT-2026-10 is scored against the five-minute configuration distribution objective, which the company set in its own service documentation. run LT-2026-10 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Warehouse connector burst after outbox partitioning run LT-2026-11, taken with the other runs in the series, shows that the test environment carried a smaller dataset than production, so the database term is understated here. 240-robot run LT-2026-11 was designed to change one variable, namely the presence of a failure injection. The target load run LT-2026-11 was measured against is the 1,500 ms p95 acknowledgement objective for the warehouse connector. scenario LT-2026-11 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Load run LT-2026-12 exercised the telemetry ingest at twice the current fleet case with 2400 robots and n/a orders an hour, on 2026-04-09. LT-2026-12 is not read as a headline number: the result is consistent with the incident record, which is the strongest corroboration available. The result recorded for run LT-2026-12 is pass, which the reviewer traced to the underlying measurement. run LT-2026-12 is treated as material to the investment case rather than as a routine engineering item.

The percentile set for twelve sites, projected 2027 fleet run LT-2026-13 reads 690 ms, 2,450 ms and 6,100 ms, with errors at 0.74%. On 2026-04-14 the team ran scenario LT-2026-13, the twelve sites, projected 2027 fleet case, at 1440 robots. LT-2026-13 shows where the system runs out of headroom: the allocator partitioning programme is the gating dependency. 1440-robot run LT-2026-13 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Scenario LT-2026-14 is recorded as pass with observation. Latency in LT-2026-14 was not applicable at p50, not applicable at p95 and not applicable at p99, with an error rate of 0.00%. Reading LT-2026-14 for the binding constraint gives drain took 71 minutes and delayed the analytics load past its window. load run LT-2026-14 was excluded from an earlier draft of TDR-2025-17 and reinstated after the reviewer obtained the supporting record.

9. Dependencies with a capacity implication

Warehouse-sim-assets was assessed for support as well as licence risk, and it carries no identified risk beyond ordinary version currency. Warehouse-sim-assets affects Simulation Harness; no other component reviewed depends on it. At OSS-030 the component is warehouse-sim-assets, pinned at 1.1.0 and used simulation scenario assets. warehouse-sim-assets is the entry that changed most between the first and second document requests behind TDR-2025-17.

Support risk on csv-export takes the form of an upstream project that is active but changes interface frequently. Attribution present is the reviewer's observation on csv-export. Public registry, signature verified is the provenance recorded for csv-export. csv-export was excluded from an earlier draft of TDR-2025-17 and reinstated after the reviewer obtained the supporting record.

Licensing for image-signing is Apache-2.0 and the copyleft column reads no. Image-signing carries the approval entry approved 2025-01-15; the reviewer looked for a corresponding decision record. Image-signing obliges the company to attribute the component in the customer-facing notice file. image-signing would be a candidate for the first post-close engineering review under any reasonable plan.

Opencv-subset carries a practical obligation: to attribute the component in the customer-facing notice file. Opencv-subset is recorded as approved 2024-07-02 in the component approval log. Opencv-subset is licensed under Apache-2.0, with copyleft exposure recorded as no. opencv-subset was checked against the company's own register, which had not been updated since the change landed.

Physics-core-fork is recorded as approved with conditions 2023-04-18 in the component approval log. Physics-core-fork carries a practical obligation: to share alike any derived asset it distributes. Tracing physics-core-fork to its source gives forked from upstream in 2023 with local patches. physics-core-fork appears twice in the working papers for TDR-2025-17 because the first answer was corrected after a follow-up request.

Http-router also carries no identified risk beyond ordinary version currency, which is a supportability question rather than a licence one. No copyleft applies to http-router, whose licence is MIT. OSS-022 covers http-router at version 1.9.4, used cloud service routing. http-router was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Tsdb-client carries MIT terms, classified no for copyleft purposes. On tsdb-client the reviewer records that attribution present in the cloud notice file. Support risk on tsdb-client takes the form of no identified risk beyond ordinary version currency. tsdb-client was one of two items where TDR-2025-17 could not obtain the underlying data within the engagement.

In addition to its terms, nav-graph-solver brings a fork that must be maintained locally for as long as it is used. Nav-graph-solver carries a practical obligation: to exclude a particular build configuration from the shipped artefact. Linking mode matters for nav-graph-solver, which is used linked into the planner as a static library. nav-graph-solver is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Pgbouncer-managed shows an approval status of approved 2024-03-05 against the company's own policy. One point about pgbouncer-managed is not resolved: attribution present. Pgbouncer-managed entered the codebase by this route: public registry, signature verified. pgbouncer-managed was cross-checked against the engineering record supplied for TDR-2025-17, and the two agree.

The operational obligation attaching to media-transcode is to share alike any derived asset it distributes. Tracing media-transcode to its source gives public registry; the build configuration determines the licence. The reviewer's note on media-transcode is that the build must exclude the GPL-configured components and the reviewer verified the current build flags. media-transcode illustrates why TDR-2025-17 separates the severity of a finding from the confidence behind it.

The terms applying to grid-render are GPL-3.0 with a commercial alternative; the copyleft classification is strong. In addition to its terms, grid-render brings a single-maintainer dependency with no succession plan. Grid-render entered the codebase by this route: public registry; the commercial licence was purchased in 2024. grid-render has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Where lidar-vendor-sdk came from matters here: supplied by the sensor vendor. OSS-018 covers lidar-vendor-sdk at version 4.1.2, used lidar driver. Lidar-vendor-sdk carries this qualification: two major versions behind current; the vendor's support statement covers the deployed version to 2027. lidar-vendor-sdk is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Compliance with the yaml-parse terms requires nothing beyond retaining the licence text with the source. Yaml-parse carries this qualification: attribution present. Yaml-parse entered the codebase by this route: public registry, signature verified. yaml-parse is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

The supply route for k8s-operator-framework is public registry, signature verified. K8s-operator-framework also carries a fork that must be maintained locally for as long as it is used, which is a supportability question rather than a licence one. The reviewer's note on k8s-operator-framework is that attribution present. k8s-operator-framework was tested rather than accepted, and the test is described in the working papers behind TDR-2025-17.

10. Security findings that constrain the capacity plan

Assessment finding SEC-2025-11 reduces to a residual risk of low. The reviewer declines to soften SEC-2025-11: the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. As at the review date assessment finding SEC-2025-11 was remediated and the closure entry reads 2025-10-30. SEC-2025-11 rests on the observation that attribute absence confirmed; no working exploit chain was demonstrated; the reviewer holds that to be strong. assessment finding SEC-2025-11 is the reason one section of TDR-2025-17 concludes narrowly rather than generally.

SEC-2026-13 against Order Broker translates into the following exposure: a large payload degrades broker throughput. Finding SEC-2026-13 is rated Low with a numeric band of 3.3 on the tester's scoring scale. Remediation status on assessment finding SEC-2026-13 is remediated; the record shows 2026-05-15. Low-severity finding SEC-2026-13 reduces to a residual risk of low. SEC-2026-13 against Order Broker is included because a reader of TDR-2025-17 should be able to disagree with the inference without repeating the work.

What SEC-2026-11 enables is this: a non-privileged process could alter a force limit parameter. For SEC-2026-11 against Gripper Controller the record shows that file permissions observed on a laboratory robot, which is strong evidence and is graded separately from severity. SEC-2026-11 leaves residual risk that the reviewer assesses as medium; the safety controller limit is the compensating control. The reviewer's position on SEC-2026-11 is that the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. Gripper Controller finding SEC-2026-11 was checked against the corresponding entry in the company's own register, and the register was current.

Moderate evidence supports SEC-2026-08, namely that role grant performed in a staging tenant. On 2026-04-18 the tester recorded Identity and Authorization finding SEC-2026-08: identity service admin role assignable without a second approval. Identity and Authorization finding SEC-2026-08 leaves residual risk that the reviewer assesses as low; role grants now require a second approver and are alerted. Remediation status on Identity and Authorization finding

SEC-2026-08 is remediated; the record shows 2026-06-30. SEC-2026-08 against Identity and Authorization is the clearest instance in TDR-2025-17 of engineering practice being better than the documentation suggests.

The reviewer's position on Medium-severity finding SEC-2026-07 is that the fix is verified and the class of defect is not, since no test now prevents its return. SEC-2026-07 leaves residual risk that the reviewer assesses as medium; network segmentation is the only current control. Moderate evidence supports SEC-2026-07, namely that operating system version observed on the rig. At 6.7 the score on assessment finding SEC-2026-07 places it in the Medium group. Medium-severity finding SEC-2026-07 is the clearest instance in TDR-2025-17 of a control that exists on paper and is weaker in practice.

Medium-severity finding SEC-2025-07 translates into the following exposure: the same secret was reusable at any site provisioned from that template. What remains after assessment finding SEC-2025-07 is remediated is medium; two forked templates were not re-issued until May 2026. Evidence behind SEC-2025-07 is that the secret was recovered from a template export, which the reviewer treats as moderate. As at the review date assessment finding SEC-2025-07 was remediated and the closure entry reads 2026-01-16. SEC-2025-07 is reported here in full because a summary of it in an earlier draft of TDR-2025-17 was misread as an assurance.

SEC-2025-09 is not fully retired by its remediation: medium; a two-person rule is designed but not enforced in the tool. Assessment finding SEC-2025-09 is partially remediated, with a closure date of pending. Finding SEC-2025-09 arrives as a Medium finding with a 6.3 score, and the reviewer did not adjust either. Weighing SEC-2025-09 against OTA Updater against the compensating controls, the fix is verified and the class of defect is not, since no test now prevents its return. SEC-2025-09 is one of the entries where TDR-2025-17 relies on the company's own measurement rather than an independent one.

In Low-severity finding SEC-2026-18 the exploitation path means a diagnostic flood can slow pack telemetry reporting. The tester scored assessment finding SEC-2026-18 at 3.4 and labelled it Low. SEC-2026-18 rests on the observation that flood produced on a bench pack; the reviewer holds that to be limited, resting on a configuration read rather than on a demonstrated exploitation. The reviewer declines to soften Battery Management Service finding SEC-2026-18: the remediation is adequate and the time to remediate is not. assessment finding SEC-2026-18 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

The tester scored assessment finding SEC-2026-16 at 5.8 and labelled it Medium. In assessment finding SEC-2026-16 the exploitation path means on-network access to a live perception diagnostic stream. SEC-2026-16 against Costmap Fusion produces this judgement: the severity is right and the ownership is unclear, since two teams each believe the other holds it. SEC-2026-16 against Costmap Fusion reduces to a residual risk of low; the socket binds to the loopback interface only. SEC-2026-16 would move from a concern to a verified defect if one further measurement were supplied.

SEC-2026-14 against Traffic Manager reduces to a residual risk of low. SEC-2026-14 against Traffic Manager arrives as a Low finding with a 3.8 score, and the reviewer did not adjust either. Remediated is the status of SEC-2026-14, dated 2026-06-18. On finding SEC-2026-14 the conclusion is that the remediation is adequate and the time to remediate is not. finding SEC-2026-14 would move from a concern to a verified defect if one further measurement were supplied.

Severity on SEC-2026-15 is Low, scored 3.0 by the assessor. Weighing finding SEC-2026-15 against the compensating controls, the retest evidence is a screenshot rather than a re-run of the original test, which the reviewer notes. Residual risk after remediation of Low-severity finding SEC-2026-15 is low; management judged the operational cost of enforcement higher than the risk. For finding SEC-2026-15 the record shows that collision created in a staging tenant, which is strong evidence and is graded separately from severity. SEC-2026-15 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

For assessment finding SEC-2025-05 the record shows that the tester submitted synthetic telemetry from an unenrolled host, which is moderate evidence and is graded separately from severity. SEC-2025-05 is not fully retired by its remediation: low; attestation is now required on the ingest path. On SEC-2025-05 the conclusion is that the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. Telemetry Ingest finding SEC-2025-05 translates into the following exposure: a forged agent could inject telemetry attributed to a real robot. SEC-2025-05 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Release Orchestrator finding SEC-2026-01 leaves residual risk that the reviewer assesses as high until the second-approver control lands. SEC-2026-01 matters because one insider can move an unreviewed artefact to production. Evidence behind Release Orchestrator finding SEC-2026-01 is that the tester used the path in a staging tenant and the audit trail recorded no justification, which the reviewer treats as strong. SEC-2026-01 was opened on 2026-04-14 and states that emergency release path permits a single approver. High-severity finding SEC-2026-01 is the clearest instance in TDR-2025-17 of a control that exists on paper and is weaker in practice.

11. Capacity qualifications and unresolved questions

5 items in TDR-2025-17 are open, two of which the reviewer would treat as pre-closing information requests rather than post-close work. Calder Rowe Engineering Advisory will not treat an open item in the report at TDR-2025-17 as closed on a verbal assurance, and has said so to the engineering leadership. Not every finding in the present report is accepted by management: 5 are declined, and the declinatures are reproduced rather than argued away. 5 declined items are carried in this report, of which two lack any documented compensating control. J. Ferreiro notes that the company answered every document request, though two answers arrived after the draft.

Where report TDR-2025-17 says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. The report at TDR-2025-17 grades evidence as strong, moderate or limited, and a limited grade is not upgraded because the consequence would be serious. The pack the present report proposes is deliberately short, because a longer one would be produced and not read. The monitoring pack the report at TDR-2025-17 proposes would have surfaced two of the incidents examined here before a customer noticed. The company reviewed a draft of the present report for factual accuracy and its corrections are reflected without altering any conclusion.

The reviewer asked in each case whether a declined item had a compensating control, and TDR-2025-17 reports the answer even where it is unsatisfactory. 5 items in TDR-2025-17 sit in a declined state, and the reviewer regards one of them as wrongly declined. The interview programme behind this report covered 7 teams and produced 14 follow-up requests, of which most were satisfied. J. Ferreiro interviewed 17 engineers and managers for the report at TDR-2025-17; two declined to be quoted on one topic and that is noted where it matters. J. Ferreiro has stated the limitations at the point they bite rather than collecting them into a disclaimer.

The present report proposes 5 metrics for the post-close monitoring pack, each already produced by an existing system. Monitoring recommended by the report at TDR-2025-17 relies on metrics the company already emits, so no new instrumentation is required to start. J. Ferreiro asked in every declined case who would revisit the decision and when, and this report records the answer even where there was none. The present report keeps declined items visible: risk acceptance without an expiry date is itself recorded as a finding. J. Ferreiro has not smoothed the sequence of this report; it follows the order in which the evidence arrived.

12. Conclusions on capacity against the operating plan

J. Ferreiro's conclusion in the report at TDR-2025-17 is that the reviewer would proceed, with the qualifications recorded in this report left on the record rather than closed out for tidiness. Stated without qualification for once, the conclusion of the report at TDR-2025-17 is that the reviewer would proceed, with the qualifications recorded in this report left on the record rather than closed out for tidiness. Sampling in TDR-2025-17 covered 21 of the 34 catalogued services and 7 of the 18 live sites. Selection for the report at TDR-2025-17 favoured components that appear in the incident record, which biases the sample towards known trouble and is stated as such. The report at TDR-2025-17 contains 5 open items, and a reader should assume that number will move rather than settle.

The transaction reading of TDR-2025-17 is that the engineering risk is priced in the plan rather than in the purchase agreement. The costed remediation implied by report TDR-2025-17 is $855,000 over the coming four quarters, which is within the ordinary engineering budget. Calder Rowe Engineering Advisory asked the same four questions of every team interviewed for TDR-2025-17, which is how the inconsistencies below were found. This report draws on 17 interviews, including the engineering owners of every component examined in detail. Each section of the present report states its own basis, so a reader can accept one section and reject another.

Calder Rowe Engineering Advisory recommends that the 5 indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. Reporting under TDR-2025-17 should distinguish

a target that was met from a target that was lowered, which the current pack does not. For the report at TDR-2025-17 the reviewer sampled 7 sites, including the largest deployment and the newest. The population for TDR-2025-17 was the service inventory as at 2025-10-24; 21 entries were examined and the rest were reviewed only at inventory level. Figures quoted in report TDR-2025-17 come from the company's systems unless the text says the reviewer measured them.

Payload Scale Interface ships as needed; the reviewer sampled the last twelve releases. Payload Scale Interface was weighed on its coverage, its defect backlog and its dependency position, and it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Payload Scale Interface rests on the interface specification, calibration records for nine units and a module-swap test so far as this review is concerned. Automated test coverage on Payload Scale Interface stands at 36 per cent line coverage. Payload Scale Interface carries a single measurable objective, namely that payload mass is reported within 2 per cent of the reference load. Payload Scale Interface is included because a reader of TDR-2025-17 should be able to disagree with the inference without repeating the work.

The published objective for Gripper Controller is that a grasp completes or reports failure inside 1.2 s. The Gripper Controller codebase is C++17 throughout, deployed onto on-robot Linux with a firmware coprocessor, and its durable record is a local parameter store. On Gripper Controller the conclusion is qualified: the reviewer accepts the engineering position while noting that the verifying test does not exist. Test coverage on Gripper Controller is 41 per cent line coverage and the measurement basis was confirmed with Ines Duarte. The payload scale interface and the perception grasp proposals feed Gripper Controller; a failure in Gripper Controller is felt by the order execution path and the safety controller fault input. Gripper Controller was tested rather than accepted, and the test is described in the working papers behind TDR-2025-17.